

Exp A Fabry-Pérot optical cavity, made of a semiconductor material ($n = 3.6$), with two mirrors separated a length of $250\mu\text{m}$. The mirrors have a reflectance ($\tilde{R}$) of 0.9

- 1.) Calculate the cavity's nearest mode when a 1310nm (near-IR) laser is used as the source.
- 2.) Calculate the separation of the modes (Δf_f) in the cavity
- 3.) Calculate the Finesse (F)
- 4.) Calculate the spectral width of each mode in the cavity
- 5.) Calculate the Q-factor.

Solution

$$m = \frac{2L}{\lambda_m} \rightarrow \lambda_m = \frac{\lambda_0}{n}$$

$$m = \frac{2nL}{\lambda_0} \quad \leftarrow m \text{ has to be an integer (Round)}$$

$$m = \frac{2(3.6)(250\mu)}{1310\text{ nm}}$$

$$m = 1374.05$$

$$m = 1374$$

Problem 2

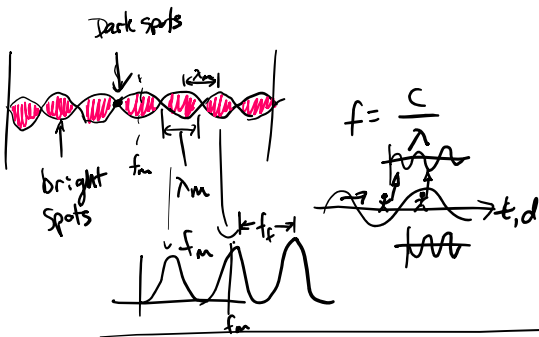
$$f_m = m f_f \quad m = \frac{2L}{\lambda_m} \quad ; \quad \lambda_m = \frac{c}{f_m}$$

$$m = \frac{2L}{c} (f_m)$$

$$m f_f = \frac{2L}{c} (m f_f)$$

$$1 = \frac{2L}{c} f_f$$

$$f_f = \frac{c}{2L} = \frac{3 \times 10^8}{2(259\mu)}$$



$$f_f = 6 \times 10^{11} \text{ Hz} \rightarrow 600 \text{ GHz}$$

Problem 3

$$F = \frac{\pi(\tilde{R})^{1/2}}{(1 - \tilde{R})} = \frac{\pi(0.9)^{1/2}}{(1 - 0.9)} = 29.8 = F$$

Problem 4

$$\text{Spectral width } \Delta f_m = \frac{f_f}{F} = \frac{6 \times 10^{11}}{29.8} = 2.01 \times 10^{10} \text{ Hz} = 20.1 \text{ GHz}$$

Quality Factor

$$Q = \frac{f_m}{\Delta f_m} = \frac{(mf_f)}{2.01 \times 10^{10} \text{ Hz}}$$

$$Q = 40945.2$$

